Algorithms

Week 3 – Arrays, Stacks, Queues, and Basic Sorting Algorithms

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Part 1. Fundamental Data Structures and Elementary Sorting

Learning Objectives

- Review fundamental data structures: list, stack, queue, heap
- Understand elementary sorting algorithms and their O(n^2) behavior
- Understand advanced sorting algorithms and their O(n log n) behavior
- Understand linear-time sorting and the conditions that enable it
- Grasp the **recursive (inductive) structure** of sorting algorithms
- Compare sorting algorithm complexities

Linked List

- A sequence of **nodes**, each containing data and a pointer to the next node

```c
typedef int element;

typedef struct ListNode {
    element data;
    struct ListNode *link;
} ListNode;
```

- Operations: insert, delete, search – all O(n) in the worst case
- Visualization: https://visualgo.net/en/list

Stack

- **LIFO** (Last In, First Out)
- `push()` : add element to the top
- `pop()` : remove element from the top

```
top →     | C |
          | B |
bottom →  | A |
```

- Applications: function call stack, expression evaluation, backtracking
- Visualization: https://visualgo.net/en/list

Queue

- **FIFO** (First In, First Out)
- `enqueue()` : add element to the rear
- `dequeue()` : remove element from the front

```
dequeue ←— [ A | B | C | D ] ←— enqueue
           front         rear
```

- Applications: BFS, scheduling, buffering
- Visualization: https://visualgo.net/en/list

Heap

- A **complete binary tree** satisfying the heap property
  - **Max Heap**: key(parent) >= key(child)
  - **Min Heap**: key(parent) <= key(child)

```
   Min Heap              Array Representation
      3                  A = [3, 6, 4, 8, 9, 7]
     / \                      1  2  3  4  5  6
    6   4
   / \   \               A[i]'s children: A[2i], A[2i+1]
  8   9   7              A[i]'s parent:   A[floor(i/2)]
```

- Heaps are stored as **arrays** (no pointers needed)
- Applications: priority queues, heap sort
- Visualization: https://visualgo.net/en/heap

Sorting Algorithms – Landscape

- Most sorting algorithms fall between **O(n^2)** and **O(n log n)**
- When input has special properties, **O(n)** sorting is possible

Category	Algorithms	Complexity
Elementary	Selection, Bubble, Insertion	O(n^2)
Advanced	Merge, Quick, Heap	O(n log n)
Linear-time	Radix, Counting	O(n)

Two perspectives on algorithms:

- **Flow-based**: follow the execution step by step
- **Relational**: observe how each step transforms the state (deeper understanding)

Elementary Sorting – O(n^2)

Selection Sort, Bubble Sort, Insertion Sort

Selection Sort – Idea

For each iteration:

1. **Find** the maximum element in the unsorted portion
2. **Swap** it with the rightmost element of the unsorted portion
3. **Exclude** that element (it is now in its final position)
4. Repeat until one element remains

```
[15, 11, 29, 20, 65, 3, 73, 48, 31, 8]   Find max (73)
[15, 11, 29, 20, 65, 3,  8, 48, 31|73]   Swap 73 with last, exclude 73
[15, 11, 29, 20,  8, 3, 11, 48, 31|65,73] Find max (65), swap, exclude
   ...
[ 3,  8, 11, 15, 20, 29, 31, 48, 65, 73]  Sorted!
```

- Selects the **maximum** (or minimum) each round

Selection Sort – Pseudocode

```
selectionSort(A[], n)          ▷ Sort A[1 … n]
{
    for last ← n downto 2 {                          — ①
        Find the largest element A[k] in A[1 … last]; — ②
        A[k] ↔ A[last];   ▷ swap A[k] and A[last]   — ③
    }
}
```

Complexity Analysis:

- Loop ① runs **n - 1** times
- Finding the max ② requires: n-1, n-2, …, 2, 1 comparisons
- Swap ③ is constant time

$$T(n) = (n-1) + (n-2) + \cdots + 2 + 1 = \frac{n(n-1)}{2} = \Theta(n^2)$$

- **Worst case**: Theta(n^2) | **Average case**: Theta(n^2)

Selection Sort – Step-by-Step Example

Step	Array State	Action
0	`[15, 11, 29, 20, 65, 3, 73, 48, 31, 8]`	Initial array
1	`[15, 11, 29, 20, 65, 3, |8|, 48, 31, |73|]`	Max=73, swap with A[10]
2	`[|8|, 11, 29, 20, |65|, 3, 15, 48, 31, 73]`	Max=65, swap with A[9]
3	`[8, 11, 29, 20, |31|, 3, 15, |48|, 65, 73]`	Max=48, swap with A[8]
…	…	Continue finding max, swapping
9	`[3, 8, 11, 15, 20, 29, 31, 48, 65, 73]`	Sorted

Animation: https://visualgo.net/en/sorting

Bubble Sort – Idea

For each iteration:

1. Starting from the left, compare **adjacent pairs**
2. If they are out of order, **swap** them
3. The largest element "**bubbles up**" to the rightmost position
4. Exclude the rightmost element and repeat

```
[15, 65, 29, 20, 11, 8, 73, 48, 31, 3]   Compare adjacent pairs
[15, 29, 20, 11, 8, 65, 48, 31, 3, |73|]  73 bubbles to end, exclude
[15, 20, 11, 8, 29, 48, 31, 3, |65, 73|]  65 bubbles, exclude
   ...
[3, 8, 11, 15, 20, 29, 31, 48, 65, 73]    Sorted!
```

- Bubbles the **maximum** to the end each round

Bubble Sort – Pseudocode

```
bubbleSort(A[], n)              ▷ Sort A[1 ... n]
{
    for last ← n downto 2                                  — ①
        for i ← 1 to last-1                                — ②
            if (A[i] > A[i+1]) then A[i] ↔ A[i+1];        — ③
}
```

Complexity Analysis:

- Loop ① runs **n - 1** times
- Loop ② runs n-1, n-2, …, 2, 1 times respectively
- Swap ③ is constant time

$$T(n) = (n-1) + (n-2) + \cdots + 2 + 1 = \frac{n(n-1)}{2} = \Theta(n^2)$$

- **Worst case**: Theta(n^2) | **Average case**: Theta(n^2)

Bubble Sort – Step-by-Step Example

Pass	Array After Pass	Bubbled Element
0	`[15, 65, 29, 20, 11, 8, 73, 48, 31, 3]`	(initial)
1	`[15, 29, 20, 11, 8, 65, 48, 31, 3, |73|]`	73
2	`[15, 20, 11, 8, 29, 48, 31, 3, |65, 73|]`	65
3	`[15, 11, 8, 20, 29, 31, 3, |48, 65, 73|]`	48
…	…	…
9	`[3, 8, 11, 15, 20, 29, 31, 48, 65, 73]`	Done

Within each pass, adjacent elements are compared left to right and swapped if out of order.

Animation: https://visualgo.net/en/sorting

Insertion Sort – Idea

For each iteration:

1. Take the next element from the unsorted portion
2. **Shift** elements in the sorted portion that are larger
3. **Insert** the element into its correct position

```
[29]  10  14  37  13          29 is trivially sorted
[10, 29]  14  37  13          Insert 10: shift 29, place 10
[10, 14, 29]  37  13          Insert 14: shift 29, place 14
[10, 14, 29, 37]  13          Insert 37: already in place
[10, 13, 14, 29, 37]          Insert 13: shift 37,29,14, place 13
```

- Inserts each element into an already **sorted** subarray

Insertion Sort – Pseudocode

```
insertionSort(A[], n)          ▷ Sort A[1 … n]
{
    for i ← 2 to n                                          — ①
        Insert A[i] into its proper place in A[1 … i];      — ②
}
```

Complexity Analysis:

- Loop ① runs **n - 1** times
- Insertion ② requires at most i - 1 comparisons

Case	Comparisons	Complexity
Worst (reverse sorted)	1 + 2 + … + (n-1)	Theta(n^2)
Average	1/2 (1 + 2 + … + (n-1))	Theta(n^2)
Best (already sorted)	1 + 1 + … + 1	**Theta(n)**

Insertion Sort – Inductive Proof of Correctness

Loop Invariant: At the start of iteration i, the subarray A[1…i-1] is sorted.

- **Base case** (n = 1): A single element A[1] is trivially sorted.
- **Inductive step**: If A[1…k] is sorted (P(k) holds), then inserting A[k+1] into its correct position yields a sorted A[1…k+1] (P(k+1) holds).
- **Conclusion**: After iteration n, the entire array A[1…n] is sorted.

> This is exactly **mathematical induction** from high school – applied to algorithms.

Recursive Structure of Elementary Sorts

All three elementary sorts have the **same recursive structure**:

```
T(n) = T(n-1) + Theta(n)
```

Algorithm	Recursive Form	"Work per level"
Insertion Sort	Sort A[1…n-1], then insert A[n]	insert(n) = Theta(n)
Selection Sort	Find max and swap, then sort A[1…n-1]	maxSwap(n) = Theta(n)
Bubble Sort	Bubble max to end, then sort A[1…n-1]	bubble(n) = Theta(n)

```
insertionSort(A, n) {
  if (n == 1) return;
  insertionSort(A, n-1);
  insert(A, n);
}
```

```
selectionSort(A, n) {
  if (n == 1) return;
  maxSwap(A, n);
  selectionSort(A, n-1);
}
```

```
bubbleSort(A, n) {
  if (n == 1) return;
  bubble(A, n);
  bubbleSort(A, n-1);
}
```

> Key difference: Insertion sort does its work **after** the recursive call; Selection and Bubble sort do their work **before**.

Part 2. Advanced Sorting – O(n log n)

Merge Sort, Quick Sort, Heap Sort

Merge Sort – Idea

Divide and Conquer:

1. **Divide**: Split the array into two halves
2. **Conquer**: Recursively sort each half
3. **Combine**: Merge the two sorted halves

```
[31, 3, 65, 73, 8, 11, 20, 29, 48, 15]    Original

[31, 3, 65, 73, 8] | [11, 20, 29, 48, 15]  ① Divide

[3, 8, 31, 65, 73] | [11, 15, 20, 29, 48]  ②③ Sort each half

[3, 8, 11, 15, 20, 29, 31, 48, 65, 73]     ④ Merge
```

Merge Sort – Pseudocode

```
mergeSort(A[], p, r)           ▷ Sort A[p ... r]
{
    if (p < r) then {
        q ← floor((p + r) / 2);       ▷ midpoint
        mergeSort(A, p, q);            ▷ sort left half
        mergeSort(A, q+1, r);          ▷ sort right half
        merge(A, p, q, r);            ▷ merge two sorted halves
    }
}
```

```
merge(A[], p, q, r)            ▷ Merge A[p ... q] and A[q+1 ... r]
{
    i ← p; j ← q+1; t ← 1;
    while (i ≤ q and j ≤ r) {
        if (A[i] ≤ A[j])
            tmp[t++] ← A[i++];
        else
            tmp[t++] ← A[j++];
    }
    Copy remaining elements to tmp[];
    Copy tmp[] back to A[p ... r];
}
```

Merge Sort – Merge Procedure Example

Merge `[3, 8, 31, 65, 73]` and `[11, 15, 20, 29, 48]`:

```
 i                        j
[3, 8, 31, 65, 73]     [11, 15, 20, 29, 48]     tmp = []

Step 1: 3 < 11  → tmp = [3],        i moves right
Step 2: 8 < 11  → tmp = [3, 8],     i moves right
Step 3: 31 > 11 → tmp = [3, 8, 11], j moves right
Step 4: 31 > 15 → tmp = [3, 8, 11, 15], j moves right
Step 5: 31 > 20 → tmp = [3, 8, 11, 15, 20], j moves right
Step 6: 31 > 29 → tmp = [3, 8, 11, 15, 20, 29], j moves right
Step 7: 31 < 48 → tmp = [3, 8, 11, 15, 20, 29, 31], i moves right
Step 8: 65 > 48 → tmp = [3, 8, 11, 15, 20, 29, 31, 48], j moves right
Step 9-10: Copy remaining [65, 73]

Result: [3, 8, 11, 15, 20, 29, 31, 48, 65, 73]
```

Merge Sort – Complexity Analysis

Recurrence:

$$T(n) = 2T\left(\frac{n}{2}\right) + \Theta(n)$$

Recursion tree: each level does Theta(n) total work, and there are log2(n) levels.

```
Level 0:          n                      → n work
Level 1:      n/2   n/2                 → n work
Level 2:    n/4 n/4 n/4 n/4             → n work
   ...
Level h:    1  1  1  ...  1             → n work

h = log₂n  levels  →  Total: n × log₂n
```

$$T(n) = \Theta(n \log n)$$

Merge sort is **always** Theta(n log n) – worst, average, and best case. Trade-off: requires **O(n) extra space** for the temporary array.

Quick Sort – Idea

1. Choose a **pivot** element (e.g., the last element)
2. **Partition**: rearrange so that elements < pivot are on the left, elements > pivot are on the right
3. The pivot is now in its **final sorted position**
4. Recursively sort the left and right subarrays

```
[31, 8, 48, 73, 11, 3, 20, 29, 65, 15]   pivot = 15

[8, 11, 3, |15|, 31, 48, 20, 29, 65, 73]  After partition

[3, 8, 11, |15|, 20, 29, 31, 48, 65, 73]  Recursively sort left & right
```

Quick Sort – Pseudocode

```
quickSort(A[], p, r)           ▷ Sort A[p … r]
{
    if (p < r) then {
        q ← partition(A, p, r);    ▷ partition around pivot
        quickSort(A, p, q-1);      ▷ sort left subarray
        quickSort(A, q+1, r);      ▷ sort right subarray
    }
}
```

```
partition(A[], p, r)
{
    pivot ← A[r];                  ▷ choose last element as pivot
    i ← p - 1;
    for j ← p to r-1 {
        if (A[j] ≤ pivot) {
            i ← i + 1;
            A[i] ↔ A[j];
        }
    }
    A[i+1] ↔ A[r];                ▷ place pivot in final position
    return i + 1;
}
```

`i` marks the boundary: everything at or below index `i` is <= pivot. `j` scans through the array from left to right.

Quick Sort – Partition Example

Partition `[31, 8, 48, 73, 11, 3, 20, 29, 65, 15]` with pivot = 15:

```
pivot = A[10] = 15,  i = 0

j=1: A[1]=31 > 15  → skip
j=2: A[2]=8  ≤ 15  → i=1, swap A[1]↔A[2] → [8, 31, 48, 73, 11, 3, 20, 29, 65, 15]
j=3: A[3]=48 > 15  → skip
j=4: A[4]=73 > 15  → skip
j=5: A[5]=11 ≤ 15  → i=2, swap A[2]↔A[5] → [8, 11, 48, 73, 31, 3, 20, 29, 65, 15]
j=6: A[6]=3  ≤ 15  → i=3, swap A[3]↔A[6] → [8, 11, 3, 73, 31, 48, 20, 29, 65, 15]
j=7: A[7]=20 > 15  → skip
j=8: A[8]=29 > 15  → skip
j=9: A[9]=65 > 15  → skip

Final: swap A[i+1]↔A[r] → swap A[4]↔A[10]
Result: [8, 11, 3, |15|, 31, 48, 20, 29, 65, 73]
                     ↑ pivot in final position (index 4)
```

Animation: https://visualgo.net/en/sorting

Quick Sort – Complexity Analysis

Recurrence: T(n) = T(i - 1) + T(n - i) + Theta(n), where i is the pivot's final position.

Case	Partition Balance	Recurrence	Complexity
Worst	0 : n-1 (sorted input)	T(n) = T(n-1) + Theta(n)	**Theta(n^2)**
Best	n/2 : n/2 (perfect split)	T(n) = 2T(n/2) + Theta(n)	**Theta(n log n)**
Average	random partition	See below	**Theta(n log n)**

Worst case – Input already sorted, pivot always min or max:

```
T(n) = T(0) + T(n-1) + Θ(n) = T(n-1) + Θ(n)
     = Θ(n) + Θ(n-1) + ... + Θ(1) = Θ(n²)
```

Quick Sort – Why Average Case is O(n log n)

Key insight: As long as the partition ratio is **any constant fraction** (even 1:9 or 1:99), the depth remains O(log n).

```
Even a 1/10 : 9/10 split:
    n
   / \
 n/10  9n/10
 / \    / \
...     ...
```

- The longest path: n -> 9n/10 -> (9/10)^2 n -> … -> 1
- Depth = log_{10/9}(n) = O(log n)
- Each level still does O(n) total work

Average-case proof (using induction):

- Assume T(i) <= c * i * log(i) for all i < n
- Average over all possible pivot positions: T(n) = (1/n) * sum_{i=0}^{n-1} [T(i) + T(n-i-1)] + Theta(n)
- By integration approximation: T(n) <= c * n * log(n)

Heap Sort – Heap Recap

Heap: A complete binary tree stored as an array, with the heap property.

```
Min Heap Example          Max Heap Example
    3                         9
   / \                       / \
  6   4                     7   8
 / \   \                   / \   \
8   9   7                 3   6   4
```

Array representation (1-indexed):

- Children of A[i]: **A[2i]** and **A[2i + 1]**
- Parent of A[i]: **A[floor(i/2)]**
- Sibling of A[i]: **A[i-1]** (when i is odd)

Heap Sort – Algorithm

```
heapSort(A[], n)                ▷ Sort A[1 ... n]
{
    buildHeap(A, n);            ▷ Build a min-heap (or max-heap)
    for i ← n downto 2 {
        A[1] ↔ A[i];           ▷ Swap root (min/max) with last
        heapify(A, 1, i-1);    ▷ Restore heap property
    }
}
```

Two main subroutines:

1. `buildHeap` : Convert an arbitrary array into a heap – **O(n)**
2. `heapify` : Fix the heap property at a single node – **O(log n)**

> **Worst case**: O(n log n) – guaranteed, unlike Quick Sort!

Heap Sort – buildHeap and heapify

```
buildHeap(A[], n)
{
    for i ← floor(n/2) downto 1       ▷ Start from last internal node
        heapify(A, i, n);
}
```

```
heapify(A[], k, n)                     ▷ Fix heap rooted at A[k]
{                                      ▷ Subtrees of A[k] already heaps
    left ← 2k;  right ← 2k + 1;
    if (right ≤ n) then {              ▷ Two children
        if (A[left] < A[right])
            smaller ← left;
        else smaller ← right;
    }
    else if (left ≤ n) then            ▷ Only left child
        smaller ← left;
    else return;                       ▷ Leaf node

    if (A[smaller] < A[k]) then {      ▷ Heap property violated
        A[k] ↔ A[smaller];
        heapify(A, smaller, n);        ▷ Recurse down
    }
}
```

Heap Sort – buildHeap Example

Build a min-heap from `A = [7, 9, 4, 8, 6, 3]`:

```
Step (a): Start          Step (b): heapify(3)    Step (c): heapify(2)
     7                        7                       7
    / \                      / \                     / \
   9   4                    9   3                   6   3
  / \   \                  / \   \                 / \   \
 8   6   3                8   6   4               8   9   4

Step (d): heapify(1)     Step (e): Final heap
     3                        3
    / \                      / \
   6   4                    6   4
  / \   \                  / \   \
 8   9   7                8   9   7

Array: [3, 6, 4, 8, 9, 7]
```

buildHeap processes nodes from **bottom to top** (floor(n/2) down to 1).

Heap Sort – Sorting Phase

After buildHeap, repeatedly extract the minimum:

```
(a) [3,6,4,8,9,7]   Swap A[1]↔A[6]: [7,6,4,8,9,|3]   heapify → [4,6,7,8,9,|3]
(b) [4,6,7,8,9,|3]  Swap A[1]↔A[5]: [9,6,7,8,|4,3]   heapify → [6,8,7,9,|4,3]
(c) [6,8,7,9,|4,3]  Swap A[1]↔A[4]: [9,8,7,|6,4,3]   heapify → [7,8,9,|6,4,3]
(d) [7,8,9,|6,4,3]  Swap A[1]↔A[3]: [9,8,|7,6,4,3]   heapify → [8,9,|7,6,4,3]
(e) [8,9,|7,6,4,3]  Swap A[1]↔A[2]: [9,|8,7,6,4,3]   Done!

Result (descending): [9, 8, 7, 6, 4, 3]
```

With a min-heap, the sorted result is in **descending** order. With a max-heap, the sorted result is in **ascending** order.

Heap Sort – Complexity

```
heapSort(A[], n)
{
    buildHeap(A, n);            → O(n)  [tighter analysis] or O(n log n)
    for i ← n downto 2 {       → n - 1 iterations
        A[1] ↔ A[i];           → O(1)
        heapify(A, 1, i-1);    → O(log n)
    }
}
```

Phase	Complexity
buildHeap	O(n)
Sorting loop	(n-1) x O(log n) = **O(n log n)**
Total	**O(n log n)**

Heap sort is **O(n log n) in the worst case** – no degenerate inputs like quick sort. It sorts **in-place** (no extra array needed, unlike merge sort).

Linear-Time Sorting – O(n)

Radix Sort and Counting Sort

Lower Bound for Comparison-Based Sorting

Theorem: Any comparison-based sorting algorithm requires **Omega(n log n)** comparisons in the worst case.

> This means Selection, Bubble, Insertion, Merge, Quick, and Heap sort **cannot** do better than O(n log n) using only comparisons.

But: If elements have **special properties**, we can bypass comparisons entirely.

Algorithm	Condition	Complexity
Radix Sort	Elements have at most k digits (k = constant)	Theta(n)
Counting Sort	Element values are in range [-O(n), O(n)]	Theta(n)

Radix Sort

Sort by each digit position, from **least significant to most significant**, using a **stable** sort.

```
radixSort(A[], n, k)           ▷ Elements have at most k digits
{
    for i ← 1 to k
        Stable-sort A[1 ... n] on the i-th digit;
}
```

Stable Sort: Elements with equal keys maintain their **original relative order**.

Radix Sort – Example

Sort: `[0123, 2154, 0222, 0004, 0283, 1560, 1061, 2150]`

```
Original     1st digit    2nd digit    3rd digit    4th digit
 0123         1560         0004         0004         0004
 2154         2150         0222         1061         0123
 0222         1061         0123         0123         0222
 0004         0222         2150         2150         0283
 0283         0123         2154         2154         1061
 1560         0283         1560         0222         1560
 1061         2154         1061         0283         2150
 2150         0004         0283         1560         2154
```

- Each column is sorted by the highlighted digit using a **stable** sort
- After processing all k = 4 digits, the array is fully sorted

$$T(n) = k \cdot \Theta(n) = \Theta(n) \quad (\text{when } k \text{ is a constant})$$

Counting Sort

Used when element values are bounded: all values in `{1, 2, ..., k}` where k = O(n).

```
countingSort(A[], B[], n)      ▷ A[1 ... n]: input, B[1 ... n]: output
{
    for i ← 1 to k
        C[i] ← 0;                     ▷ Initialize counts

    for j ← 1 to n
        C[A[j]]++;                     ▷ Count occurrences

    // C[i] now = number of elements equal to i

    for i ← 2 to k
        C[i] ← C[i] + C[i-1];        ▷ Cumulative counts

    // C[i] now = number of elements ≤ i

    for j ← n downto 1 {
        B[C[A[j]]] ← A[j];           ▷ Place element
        C[A[j]]--;                     ▷ Decrement count
    }
}
```

Counting Sort – Example

Sort `A = [3, 1, 2, 1, 1, 4, 2, 3, 1, 2]` with k = 4:

```
Step 1: Count occurrences          Step 2: Cumulative counts
C = [4, 3, 2, 1]                  C = [4, 7, 9, 10]
     1  2  3  4                        1  2  3  4

     ↑                                 ↑
     4 ones, 3 twos,                   Elements ≤ 1: 4
     2 threes, 1 four                  Elements ≤ 2: 7
                                       Elements ≤ 3: 9
                                       Elements ≤ 4: 10

Step 3: Place elements (right to left for stability)
j=10: A[10]=2, C[2]=7 → B[7]=2, C[2]=6
j=9:  A[9]=1,  C[1]=4 → B[4]=1, C[1]=3
j=8:  A[8]=3,  C[3]=9 → B[9]=3, C[3]=8
   ...

B = [1, 1, 1, 1, 2, 2, 2, 3, 3, 4]     ← Sorted!
```

$$T(n) = \Theta(n + k) = \Theta(n) \quad (\text{when } k = O(n))$$

Complexity Comparison – All Sorting Algorithms

Algorithm	Worst Case	Average Case	Best Case	Space	Stable?
Selection Sort	Theta(n^2)	Theta(n^2)	Theta(n^2)	O(1)	No
Bubble Sort	Theta(n^2)	Theta(n^2)	Theta(n)	O(1)	Yes
Insertion Sort	Theta(n^2)	Theta(n^2)	**Theta(n)**	O(1)	Yes
Merge Sort	Theta(n log n)	Theta(n log n)	Theta(n log n)	**O(n)**	Yes
Quick Sort	**Theta(n^2)**	Theta(n log n)	Theta(n log n)	O(log n)	No
Heap Sort	Theta(n log n)	Theta(n log n)	Theta(n log n)	O(1)	No
Counting Sort	Theta(n + k)	Theta(n + k)	Theta(n + k)	O(k)	Yes

Summary

- **Data Structures**: List, Stack (LIFO), Queue (FIFO), Heap (complete binary tree)
- **Elementary Sorts** – O(n^2): Selection, Bubble, Insertion
  - All share the recursive structure T(n) = T(n-1) + Theta(n)
  - Insertion sort is best for nearly-sorted data: O(n) best case
- **Advanced Sorts** – O(n log n): Merge, Quick, Heap
  - Merge sort: always O(n log n), but needs O(n) extra space
  - Quick sort: O(n log n) average, O(n^2) worst case
  - Heap sort: O(n log n) worst case, in-place
- **Linear-time Sorts** – O(n): Radix (digit-by-digit, stable), Counting (value counts)
  - Require special input conditions to achieve linear time
- **Next week**: More advanced algorithm paradigms

Q & A

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